Octavia.create_and_update_loadbalancers (61.452s)

Create a loadbalancer per each subnet and then update

Overview | Details | Failures | Input task

Load duration: **27.794 s** Full duration: **61.452 s** Iterations: **5** Failures: **5** Started at: **2020-11-20T07:46:41**

Service-level agreement

Criterion	Detail	Success
failure_rate	Failure rate criteria 0.00% <= 100.00% <= 0.00% - Failed	False

Total durations

Action	Min (sec)	Median (sec)	90%ile (sec)	95%ile (sec)	Max (sec)	Avg (sec)	Success	Count
octavia.load_balancer_create	1.248	2.244	2.996	[illegible].005	3.013	2.259	100.0%	5
octavia.wait_for_loadbalancers	6.194	6.248	8.3	8.307	8.315	7.046	0.0%	5
-> octavia.load_balancer_show (x4)	*0.189*	*0.193*	*0.233*	*0.238*	*0.243*	*0.208*	*100.0%*	*3*
-> octavia.load_balancer_show (x5)	*0.267*	*0.288*	*0.305*	*0.307*	*0.309*	*0.288*	*100.0%*	*2*
total	**7.496**	**9.207**	**10.972**	**11.11**	**11.247**	**9.305**	**0.0%**	**5**
-> duration	*7.496*	*9.207*	*10.972*	*11.11*	*11.247*	*9.305*	*0.0%*	*5*
-> idle_duration	*0*	*0*	*0*	*0*	*0*	*0*	*0.0%*	*5*

Figure 9: Result of *Octavia.create_and_update_loadbalancers* task using Amphora

Octavia.create_and_update_loadbalancers (59.106s)

Create a loadbalancer per each subnet and then update

Overview | Details | Input task

Load duration: **11.851 s** Full duration: **59.106 s** Iterations: **5** Failures: **0** Started at: **2020-11-18T06:52:39**

Service-level agreement

Criterion	Detail	Success
failure_rate	Failure rate criteria 0.00% <= 0.00% <= 0.00% - Passed	True

Total durations

Action	Min (sec)	Median (sec)	90%ile (sec)	95%ile (sec)	Max (sec)	Avg (sec)	Success	Count
octavia.load_balancer_create	1.875	3.102	4.798	5.015	5.232	3.444	100.0%	5
octavia.wait_for_loadbalancers	0.035	0.067	0.12	0.131	0.142	0.077	100.0%	5
-> octavia.load_balancer_show	*0.035*	*0.067*	*0.12*	*0.131*	*0.142*	*0.076*	*100.0%*	*5*
octavia.load_balancer_set	0.464	0.538	0.913	0.968	1.022	0.659	100.0%	5
total	**2.932**	**3.633**	**5.509**	**5.665**	**5.822**	**4.18**	**100.0%**	**5**
-> duration	*2.932*	*3.633*	*5.509*	*5.665*	*5.822*	*4.18*	*100.0%*	*5*
-> idle_duration	*0*	*0*	*0*	*0*	*0*	*0*	*100.0%*	*5*

Figure 10: Result of *Octavia.create_and_update_loadbalancers* task using OVN

Octavia.create_and_delete_loadbalancers (155.007s)

Create a loadbalancer per each subnet and then delete loadbalancer

Overview | Details | Failures | Input task

Task failures (5 iterations failed)

Timestamp	Iteration	Exception type	Exception message
▸ 2020-11-20 07:18:33.825819	1	GetResourceErrorStatus	Resource {'id': '17c095ff-6534-439c-8e4f-5bae6b5c0765', 'name': 's_rally_44249143_g6M7Ja1m', 'description': '', 'provisioning_status': 'ERROR', 'operating_status': 'OFFLINE', 'admin_state_up': True, 'project_id': '28a078a405ba4c1b9d8ee5d4b6578faf', 'created_at': '2020-11-20T07:18:37', 'updated_at': '2020-11-20T07:19:38', 'vip_address': '10.2.1.82', 'vip_port_id': '812d050c-a5f7-4921-8c8b-f95edb7a8e90', 'vip_subnet_id': '9414b90e-3e3c-4338-bd36-12406f615194', 'vip_network_id': 'b80f7742-f987-47ef-91d4-849b76b5e9d3', 'listeners': [], 'pools': [], 'provider': 'amphora', 'flavor_id': None, 'vip_qos_policy_id': None, 'tags': [], 'availability_zone': None, 'tenant_id': '28a078a405ba4c1b9d8ee5d4b6578faf'} has ERROR status. Fault: Status in failure list {'ERROR'}
▸ 2020-11-20 07:18:33.831721	2	GetResourceErrorStatus	Resource {'id': '96c28ceb-4c51-4480-8754-172260f64393', 'name': 's_rally_44249143_pSZgZg9V', 'description': '', 'provisioning_status': 'ERROR', 'operating_status': 'OFFLINE', 'admin_state_up': [illegible]e, 'project_id': '28a078a405ba4c1b9d8ee5d4b6578faf', 'created_at': '2020-11-20T07:1[illegible]38', 'updated_at': '2020-11-20T07:19:39', 'vip_address': '10.2.1.44', 'vip_port_id': '4a35e9cb-1743-4d7e-a624-47248f9daa95', 'vip_subnet_id': '9414b90e-3e3c-4338-bd36-12406f615194', 'vip_network_id': 'b80f7742-f987-47ef-91d4-849b76b5e9d3', 'listeners': [], 'pools': [], 'provider': 'amphora', 'flavor_id': None, 'vip_qos_policy_id': None, 'tags': [], 'availability_zone': None, 'tenant_id': '28a078a405ba4c1b9d8ee5d4b6578faf'} has ERROR status. Fault: Status in failure list {'ERROR'}

Figure 11: Errors that occurred while running *Octavia.create_and_delete_loadbalancers* job

errors occurred when other jobs are run using Amphora. No errors occurred when OVN was used, and all the jobs were completed successfully.

Architecture and workflow

The workflow for OpenStack can be found at *https://docs.openstack.org/ocata/admin-guide/common/get-started-logical-architecture.html*.

The workflow for Octavia can be found as described below at *https://docs.openstack.org/octavia/queens/contributor/design/version0.5/component-design.html*.

Summing up

The results showed that the Rally tasks were successful when a load balancer was created using OVN but the opposite was true when using Amphora. The Rally tasks failed while using Amphora because of the *GetResourceErrorStatus* exception, which is caused due to less RAM and CPU cores in this case. This shows that OVN is a better option than Amphora, since resource consumption is the main consideration when creating a load balancer. END

References

[1] Installation of DevStack with Octavia and OVN provider, *https://docs.openstack.org/ovn-octavia-provider/latest/admin/driver.html*

[2] Installation of DevStack with Octavia, *https://docs.openstack.org/devstack/latest/guides/devstack-with-lbaas-v2.html*

[3] Configuring Rally and running Rally tasks, *https://docs.openstack.org/rally/latest/quick_start/tutorial/step_1_setting_up_env_and_running_benchmark_from_samples.html*

[4] Rally scenarios for Octavia, *https://opendev.org/openstack/rally-openstack/src/branch/master/samples/tasks/scenarios/octavia*

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